

seam-coverage-measurement

BOB-087 — §11.4.106(F) three-seam coverage measurement

Item: BOB-087 (RD2-20) — Wire docs_chain / commit-seam sync hook per §11.4.106(F) **Type:** Task · **Status after this measurement:** In progress (2 of 3 seams covered) **Measured:** 2026-08-21

What §11.4.106(F) requires

“the doc/DB sync is enforced at the COMMIT seam (hook refuses a commit whose staged set touches a chain source while its exports/DB are stale) + the BUILD seam (pre-build gate) + the CONSTITUTION-PULL seam (§11.4.164 post-update hook)”

Three seams. Each was measured separately.

SEAM 1 — COMMIT: COVERED (measured, both polarities)

scripts/hooks/docs-sync-commit-seam.sh exists (368 lines) and is invoked from scripts/commit-push-all.sh at **both** commit call sites — the --scope branch (line 378) and the git add -A branch (line 386) — through _docs_sync_seam_check (line 275), which runs *after* staging and *before* git commit, and exit 1s the whole wrapper on refusal. There is no third path to git commit in that script.

Live run over the tracker file set:

```
$ bash scripts/hooks/docs-sync-commit-seam.sh --files docs/
Issues.md docs/Fixed.md docs/workable_items.db
CHECK 1 workable-items validate ..... PASS
```

```
CHECK 2 workable-items diff ..... PASS
CHECK 3 body_md drift oracle ..... PASS
```

The item's literal claim, tested in both directions

The acceptance text is “*a docs/workable_items.db write can never again land without its MD mirror in the same commit*”. A one-directional oracle would satisfy the words and miss half the defect, so both directions were driven. Everything ran on temp copies; the real docs/workable_items.db sha256 was unchanged before and after (28e12f219590e667).

MD drifts, DB stale — the hook's own §11.4.107(10) self-test:

```
golden-good      PASS  (clean tree reproduces tracked DB; oracle
does not false-fire)
golden-bad       PASS  (seeded body drift DETECTED and named:
BOB-008)
```

DB written, MD stale — the direction the item actually names. A *real engine* write (workable-items update --id BOB-084 -status "In progress") was applied to a temp copy of the DB with the working-tree Markdown deliberately left alone:

```
negative control  PASS  (no DB write -> oracle silent; no false
positive §11.4.201(1))
golden-bad       PASS  (DB-only write DETECTED and named:
BOB-084)
```

So the seam refuses a DB write whose MD mirror is absent, and does not refuse a clean tree. It has also refused live commits in anger today on checks 1, 2, 3 and 5 (operator-reported).

SEAM 2 — BUILD: COVERED (with one named residual)

scripts/pre_build_verification.sh invariant 17 (CM-WORKABLE-ITEMS-VALIDATE) runs **both** halves, and passes the Markdown paths explicitly — not the flagless form:

```
workable-items validate --db "${WORKABLE_DB}"
workable-items diff --db "${WORKABLE_DB}" --issues "${ISSUES_MD}"
--fixed "${FIXED_MD}"
```

Invariant 18 / 22 cover `workable-items-export.sh` (the `.md→exports` leg, including a real-invocation assertion so a silently-skipping export cannot pass), and invariant 24 (`CM-DOCS-CHAIN-ENGINE-VERIFY`) runs the real Docs Chain engine `verify --all` against `.docs_chain/contexts/*`.

Residual (named, not silently absorbed): the build seam has no equivalent of the commit seam's CHECK 3. `diff` is structurally blind to body-only drift — that blindness is BOB-136, and it is exactly how BOB-008's Evidence text rotted unnoticed. The build seam therefore enforces doc/DB sync but cannot see the `body_md` class. This is a *narrower* instrument at that seam, not an absent one.

SEAM 3 — CONSTITUTION-PULL: NOT COVERED (measured absence, control-needed)

Neither hook that runs on a constitution pull performs any doc/DB sync check.

```
$ grep -cE "workable-items|docs-sync|docs_chain|11\.4\.106"
constitution/scripts/post_update_hook.sh
0
$ grep -cE "workable-items|docs-sync|docs_chain|11\.4\.106"
scripts/verify-all-constitution-rules.sh
0
```

A zero is not evidence until the instrument is proven able to see through the same path (§11.4.201(7)(b)). Control needles, same `grep`, same files:

<code>post_update_hook.sh</code>	<code>"skill"</code>	<code>-></code>
38 (sees)		
<code>post_update_hook.sh</code>	<code>"ZZZ_NOT_PRESENT_NEEDLE"</code>	<code>-></code>
0 (negative control)		
<code>verify-all-constitution-rules</code>	<code>"covenant_propagation_suite"</code>	<code>-></code>
7 (sees)		

The instrument sees. The zeros are real.

Nor does the sweep pick a sync check up transitively: of the 172 gates under `constitution/scripts/gates/`, only two mention the `workable-items` or `docs_chain` engines (`cm_covenant_114_202_propagation.sh`, `cm_covenant_114_213_propagation.sh`) and both are *anchor-literal*

presence gates — they assert a governance file contains a string, and perform no DB↔Markdown comparison. `config/constitution-sweep.conf` adds no such check either.

Consequence: a constitution pull that lands a doc-affecting change can be treated as canonical with the tracker's DB and Markdown never re-compared. §11.4.32 names `scripts/verify-all-constitution-rules.sh` as boba's sweep, so that is where the missing check belongs.

Remaining work (precise)

1. Add a doc/DB sync stage to `scripts/verify-all-constitution-rules.sh` that invokes the already-existing seam by reference — `bash scripts/hooks/docs-sync-commit-seam.sh --files docs/Issues.md docs/Fixed.md docs/workable_items.db` — reporting PASS / FAIL / SKIP-with-reason in the sweep's existing verdict vocabulary, never a silent pass on an absent tool (§11.4.3 / §11.4.201). No new gate logic: the seam already exists and is self-validated, so this is a wiring change, not a second implementation (§11.4.227).
2. Optional, tracked separately as the BOB-136 class: give the build seam the CHECK 3 `body_md` oracle so `diff`'s blindness is not the build seam's blindness.

Not done in this round: both files are outside the declared file scope of the working brief (`scripts/pre_build_verification.sh` is explicitly forbidden; `scripts/verify-all-constitution-rules.sh` is not in the granted scope). Naming the gap and leaving the item open is the honest outcome rather than exceeding scope (§11.4.6).

Honest boundary (§11.4.6)

The commit seam proves the tracked DB and the working-tree Markdown agree at commit time. It does not prove either side is *correct*, and it only fires when the staged set touches a chain source — a pure-code commit stages none, which is why CHECK 5 (BOB-136 closure seam) deliberately runs before the early exit.